

BIO-RAD

Model 1710 Gradient Monitor
Instruction Manual

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Model _____

Catalog Number _____

Date of Delivery _____

Warranty Period _____

Serial Number _____

WARRANTY

Bio-Rad Laboratories' Model 1710 Gradient Monitor is warranted against defects in materials and workmanship for 1 year. If any defects occur in the instrument during this warranty period, Bio-Rad Laboratories will repair or replace the defective parts free. The following defects, however, are specifically excluded:

1. Defects caused by improper operation.
2. Repair or modification done by anyone other than Bio-Rad Laboratories or an authorized agent.
3. Use of fittings or other spare parts supplied by anyone other than Bio-Rad Laboratories.
4. Damage caused by accident or misuse.
5. Damage caused by disaster.
6. Corrosion due to use of improper solvent or sample.

This warranty does not apply to parts listed below:

1. Fuses.

For any inquiry or request for repair service, contact Bio-Rad Laboratories after confirming the model and serial number of your instrument.

Section 1

General Information

1.1 Introduction

The Model 1710 Gradient Monitor is a conductivity detector for HRLC® (high resolution liquid chromatography) detection. The monitor furnishes valuable information about the small changes in ionic strength that affect a chromatographic separation. Separation results can be reported more accurately as a function of conductivity, rather than as a function of the solvent method gradient composition signal. Conductivity is measured in five ranges, from 19.99 mSFS (millisiemens full scale) to 1.999 μ SFS (microsiemens full scale).

Either Standard or High Sensitivity Flow Cells may be used with the monitor, to match the sensitivity range required in the chromatographic separation. The High Sensitivity Flow Cell is suitable for ion chromatography applications. The Standard Flow Cell is used to follow the progress of gradient chromatography. The wetted materials of these two flow cells are corrosion resistant titanium and PTFE. A dip probe is available for use in batch operations to verify the quality of deionized water, the ionic strength of buffers, and the completeness of desalting or dialysis procedures. The flow cells and dip probe are purchased separately.

The monitor features microprocessor control. In all operating modes, the chromatographer is aided by liquid crystal display messages generated by the microprocessor. The three operating modes are conductivity, concentration, and gradient percent. In the conductivity mode, the monitor measures direct conductivity in units of Siemens (S). In the concentration and gradient percent modes, the user may calibrate the output of the monitor against two values of concentration or gradient buffer composition. The monitor memory will retain the calibration values, and the microprocessor will select the appropriate sensitivity range to optimize detection. In these two calibration modes, output is based on conductivity detection.

1.2 Principle of Operation

Conductivity is a universal method for the detection of compounds that bear a charge. The conductance of an electrolytic solution is most dependent upon the concentration of the charged solutes. Conductivity may be defined as the conductance of electricity across two electrodes in an electrolytic solution. The units of conductivity are ohms⁻¹, mhos, and Siemens. The three units are equivalent.

Resistance is the reciprocal of conductance. Ohm's Law, $V=IR$, relates resistance, R , to electrical potential. The changes in electrical potential due to changes in resistance of a solution are measured at the detection cell. Resistance is a function of the number of ions, ion charge, ion size, ion interactions, temperature, and dielectric constant of the medium. Resistance decreases, and thus conductance increases, as the concentration of electrolytes increases.

At higher solute concentrations, the increase in conductance with the increase in concentration deviates from linearity. As the concentration of electrolytes increases, Coulombic interactions between adjacent ions increase, and non-conducting ion pairs are formed. Also, the mobility of a central ion and its associated oppositely charged ion cloud becomes impaired when ionic interactions in a more viscous, concentrated solution become more prevalent. In a less viscous solution, conductivity will be linear in relation to concentration.

Conductance is also sensitive to temperature, due to an exponential decrease in viscosity with increased temperature. Conductivity will increase by approximately 2% per 1° C. The Model 1710 Gradient Monitor compensates for temperature deviation at this rate from a reference of 25° C.

At the flow cell electrode surface, ions of opposite charge to the electrode will be attracted. A double layer of ions can form that increases the capacitance of the electrode, which lowers the effective potential applied to the fluid. The application of a sinusoidal wave potential eliminates the effects of capacitance formation.

Each detection cell has a different effective surface area and distance between electrodes. Specific conductivity, with units of mhos/cm or S/cm, takes this into account. To obtain units of specific conductivity, conductivity is multiplied by the cell constant, k . The cell constant is the ratio of length between electrodes to electrode surface area, and has units of cm^{-1} . To determine the true cell constant of a detection cell, a standard of known specific conductivity is used. A 1,000 $\mu\text{mho/cm}$ Conductivity Standard is available (catalog number 167-0445).

1.3 Specifications

Conductivity ranges	0 to 1.999 μS 0 to 19.99 μS 0 to 199.9 μS 0 to 1999 μS 0 to 19.99 mS
Operational modes	Conductivity, in units of microsiemens (μS) or millisiemens (mS) Concentration (calibrated mode of unspecified units) Gradient percent (calibrated mode of %B composition)
Front membrane panel	10-key pad plus decimal, CLR , ENTER , RANGE UP/DOWN , CALIB , COND , CONC , GRAD function keys
LCD message display	2 lines of 16 characters each, 5 x 8 dot matrix
LED display	7 segments, 3-1/2 digits, floating decimal point
Time constant	3 ± 1 second
Signal output	10 mV full scale and 1 V full scale
Accuracy	1% of full scale (Δ temperature $< 5^\circ \text{C}$, within the range 0 to 40°C)
Reproducibility	0.5% of full scale (0.745 g/L KCl, constant temperature, standard flow cell)
Linearity	0.5% of full scale (0 to 0.745 g/L KCl, constant temperature, standard flow cell)
Detectability	0.2% of full scale
Temperature compensation	Set to 25°C at a rate of 2% per $\Delta 1^\circ \text{C}$
Cell designs	Standard flow cell ($k = 35 \text{ cm}^{-1}$, 4 μL volume) High sensitivity flow cell ($k = 0.1 \text{ cm}^{-1}$, 35 μL volume) Dip probe ($k = 10 \text{ cm}^{-1}$)
Wetted materials	Titanium and PTFE (flow cells), gold and epoxy (dip probe)
Operating pH range	2 to 12
Operating temperature range	5 to 40°C ; 0 to 95% RH
Dimensions	280 x 120 x 160 mm
Weight	3.2 kg
Host communication	All front panel keys supported in RS-232-C communication protocol
Memory protection	Internal battery
Power requirements	90 to 132 VAC or 198 to 264 VAC, 47 to 63 Hz

Section 2

Installation and Operation

2.1 Unpacking

Take the unit out of the box, holding the bottom of the cabinet. Record the serial number printed on the serial number label. Confirm that the instrument line voltage and frequency specified on the serial number label are compatible with the line power to be used. If you find any discrepancy, contact Bio-Rad Laboratories.

The monitor is shipped with two 1 amp fuses installed. Two 1 amp spare fuses are supplied as replacements.

The external cells (standard flow cell, high sensitivity flow cell, and dip probe) are not supplied with the Model 1710 Gradient Monitor. They may be purchased separately. See Section 4 for ordering information.

2.2 Installation

Connect the cell cable of one of the three external cells (Standard, High Sensitivity, Dip Probe) to the cell connection port on the back panel of the monitor. If no device is connected, the keypad will remain locked.

For flow cell installation, connect the 1/16" OD PTFE inlet tubing of the cell to the chromatograph. Either standard HPLC stainless steel nut and ferrules or polymeric finger-tight fittings may be used. If columns from a number of manufacturers are used, we recommend that a low dead volume union be used as an adaptor to minimize dead volume and insure a secure connection. A low dead volume union will be required to connect the Model 1710 Gradient Monitor to another detector, such as a UV monitor. In general, the gradient monitor should be placed in line after such a detector. When the flow cell is connected, it may be useful to orient the cell so that there is upward flow through it.

Connect the signal binding post terminals on the rear panel of the detector to the input terminals of a chart recorder or integration device. Refer to the instruction manual for the chart recorder or integration device for information concerning the connection and operation of that instrument. For operation with a chart recorder, a full scale output of 10 mV may be selected on the monitor rear panel. Prior to operation with a chart recorder, switch the input range of the recorder to 0 V input (or the equivalent) and adjust the recorder pen position to 0% of chart. For operation with an integration device, the 1 V full scale output may be selected. If the integration device has the capability, select the baseline zero value on that device.

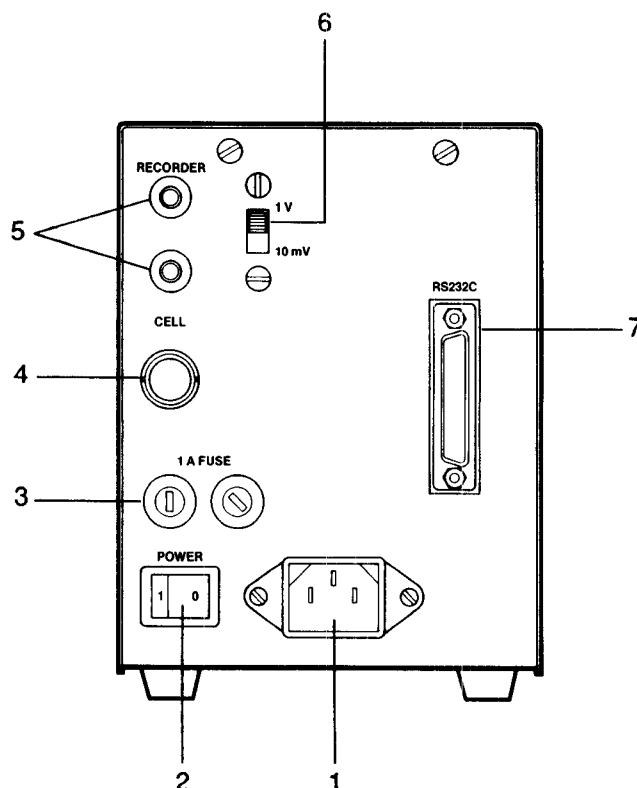
Before connecting the power cable, confirm that the switch on the rear panel is off. Plug the power cable into the power receptacle on the rear panel of the detector, then insert the power cable plug into a power outlet. The line frequency may be either 50 or 60 Hz. It is recommended that a three-prong outlet with ground be used.

Turn on the power and initiate the warm-up sequence.

Warning: Many HPLC solvents, including buffer solutions, are electrically conductive. Due caution should be exercised.

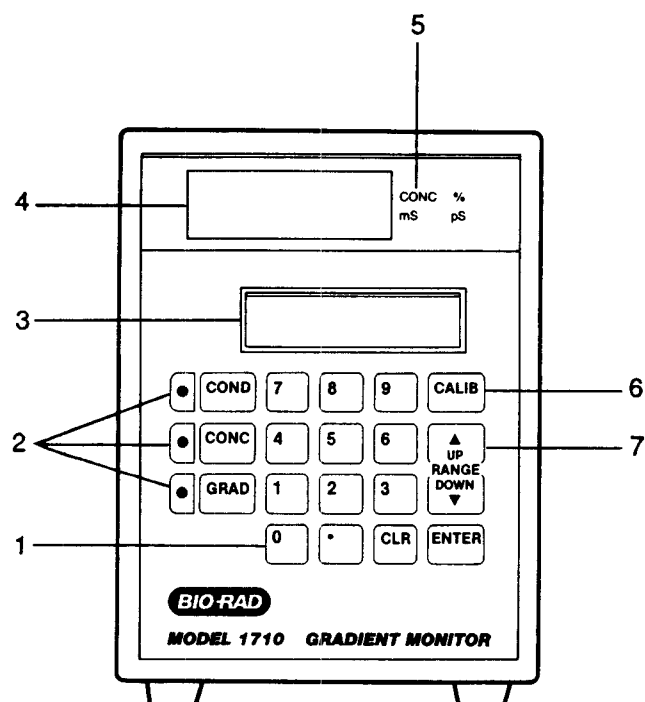
2.3 Description of Parts and Functions

Figure 2.1. Rear Panel



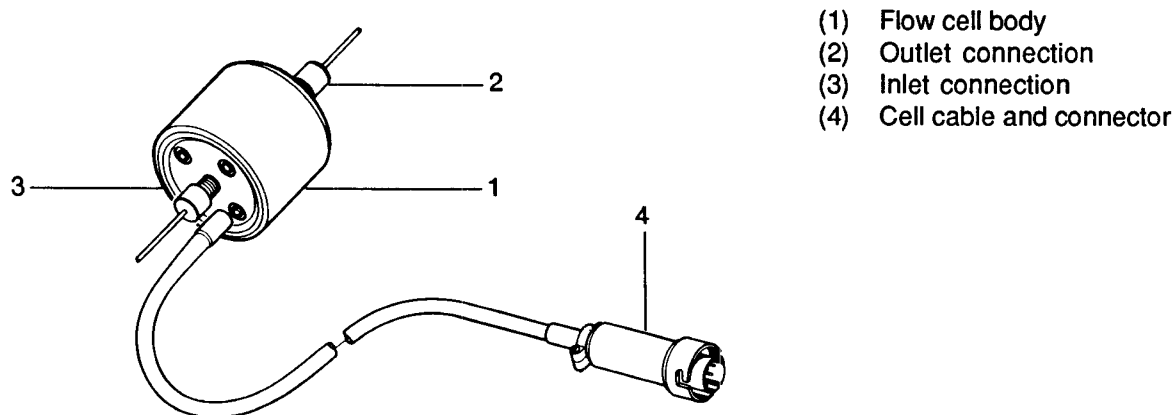
- | | |
|------------------------------|--|
| (1) Power receptacle | The power cord is connected here. |
| (2) Power switch | The power is turned on by pressing the left side (1) of the switch, and turned off by pressing the right side (0). On power-up, the instrument requires 3 minutes to warm up before control can be obtained. |
| (3) Fuse holders | Two 1 amp fuses are installed. |
| (4) External cell connection | The external cell (Standard Flow Cell, High Sensitivity Flow Cell, or Dip Probe) is connected here. |
| (5) Signal binding posts | The signal cable is connected between these terminals and an output device, such as a chart recorder or integrator. |
| (6) Signal output switch | The monitor output signal may be set at 10 mV or 1 V. |
| (7) RS-232-C connection | RS-232-C port for serial communication with supported external devices. |

Figure 2.2. Front Panel



- | | | |
|-----|------------------------|--|
| (1) | 10-key pad | The membrane panel includes a 10-key numerical pad, with ENTER and CLEAR keys. |
| (2) | Operating mode LEDs | These LEDs are adjacent to the mode keys and designate the current operating mode. |
| (3) | LCD message display | This liquid crystal display presents current status and prompts. In the conductivity mode, the current range is displayed. |
| (4) | LED display | This numerical LED display reads the current numerical output of the monitor. In the conductivity mode, the units are Siemens. In the concentration and gradient percent modes, the value is determined by the user and is not specified by the monitor. |
| (5) | Units LED | These LEDs light to indicate the units, or type of units, of the LED numerical display. |
| (6) | Calibration mode key | The calibration mode key (abbreviated CALIB) is used to initiate the procedure for the calibration of two buffers. The instrument must be in either the concentration or gradient percent mode for the calibration mode key to be active. |
| (7) | Conductivity range key | This key is used to select one of the five conductivity ranges. The key is active in only the conductivity mode. In the concentration and gradient percent modes, the optimal range is automatically selected. |

Figure 2.3. Standard Flow Cell



2.4 Operation in the Conductivity Mode

After the monitor completes its warm-up cycle on power-up, it defaults to the COND (conductivity) mode and activates the three mode keys and the two range keys. The numerical LED display shows the current conductivity at the flow cell, and the LCD displays the current conductivity range.

The range setting displayed on the LCD refers to the full scale output to a chart recorder or integration device. As an example, consider the response of a chart recorder set to a 10 mV input, if the monitor signal output and range setting are set to 10 mV and 0 to 19.99 μ S, respectively. A conductivity value of 10.0 μ S would deflect the recorder pen 50% of full scale. If the range setting were changed to 0 to 199.9 μ S, the recorder pen would deflect to 5% of full scale. At a more sensitive range setting of 0 to 1.999 μ S, an Out Of Range message would be displayed, because the 10 μ S value exceeds the range.

To change the range to a more sensitive setting, press the Range Down key. The more sensitive ranges are typically used in conjunction with the High Sensitivity Flow Cell for ion chromatography applications or high purity water measurement. To select a wider range, press the Range Up key. The wider ranges are suited to gradient monitoring with the Standard Flow Cell.

Note: There is an approximately 3 second delay in the monitor's response to a change in conductivity.

Table 2.1 Summary of Conductivity Mode Operation

Step	Operation	Notes	LCD Message	LED Display
1	POWER ON	3 minutes warming up. No key input can be accepted.	Warming up	
2		Conductance mode LED will illuminate. Range will default to 0.00 to 19.99 mS. The mS LED will light. Display range corresponds to chart recorder full scale output. In 19.99 mS range, 10 mS = 5 mV (for 10 mV recorder output).	Conductance 0-19.99 mS	Conductivity measurement
3	RANGE DOWN twice	The μ S LED will light.	Conductance 0-19.99 mS	Conductivity measurement

2.5 Operation in the Concentration Mode

In the concentration (CONC) mode, the user may enter the numerical values that correspond with any unit of measure. Range of the permissible calibration values is 0.000 to 1999. For example, the value of a 0.5 M salt may be entered as 0.5 (molar) or 500 (millimolar). The CONC mode is calibrated against two buffers in a two step process by the user. Of course, the units of the two calibrants must be the same.

To perform the CONC calibration, press CONC, then CALIB. When prompted, fill the flow cell with the first buffer (usually the low ionic strength solution). To enter the first calibrant, select the buffer A concentration value (for example, 0.5), and press ENTER. After the microprocessor has calculated the first calibration point, the LCD will prompt for the second buffer. Fill the cell with more than 2 milliliters of buffer B. Enter the second calibrant in the manner of the first.

The concentration calibration values are retained memory separate from the gradient percent calibration values. The memory is backed-up by a battery, so memory is retained after the instrument is turned off.

The user may select either of the two other operational modes, once the calibration sequence is completed. To continue operation in the CONC mode, the CONC LED must be lit.

Table 2.2. Summary of Concentration Mode Operation

Step	Operation	Notes	LCD Message	LED Display
1	CONC	Concentration LED will illuminate Previous calibration values are on LCD.	CALIB values are A=5.0 B=80.0	Concentration based on previous calibration.
2	CALIB	Operator must fill cell with buffer A.	Buffer A in cell ENTER conc value	The previous value of A is displayed.
2A	CLR	Press CLR once to clear LED value. Press CLR one more time to exit the calibration mode.		
3	5.0	New value may be entered.		5.0 flashes
4	ENTER	If the value for A is acceptable, it may be entered. Upon completion of calculation, next message appears.	A Calib in progress Buffer B in cell ENTER conc value	The previous value of B is displayed.
4A	CLR	Press CLR once to clear LED value. Press CLR one more time to exit the calibration mode.		
5	95.0	If new value for B is required, it may be entered.	B calib in progress	95.0 flashes
6	ENTER	The A value will correspond to 0 volt output on chart recorder; The B value will correspond to 100% full scale output.	CALIB values are A=5.0 B=95.0	Concentration in cell at time.

2.6 Operation in the Gradient Percent Mode

In the gradient percent (GRAD) mode, the numerical LED value corresponds to the percent of full scale deflection on the output device. GRAD is calibrated against two buffers in a two step process. The range of the permissible calibration values is 0 to 100.

To perform GRAD calibration, press GRAD, then CALIB. When prompted, fill the flow cell with the first buffer (usually the low ionic strength solution). To enter the first calibrant, select the buffer A gradient percent value (for example, 0), then press ENTER. After the microprocessor has calculated the first calibration point, the LCD will prompt for the second buffer. Fill the cell with more than 2 milliliters of buffer B. Enter the second calibrant in the manner of the first.

The gradient percent calibration values are retained memory separate from the concentration calibration values. The memory is backed up by a battery, so memory is retained after the instrument is turned off.

The user may select either of the two other operational modes, once the calibration sequence is completed. To continue operation in GRAD, GRAD LED must be lit.

Table 2.3. Summary of Gradient Percent Mode Operation

Step	Operation	Notes	LCD Message	LED Display
1	GRAD	Concentration mode LED will illuminate. GRAD LED will light. Previous calibration values are on LCD.	CALIB values are A=5.0 B=80.0	Gradient percent based on previous calibration.
2	CALIB	Fill cell with buffer A.	Buffer A in cell ENTER grad value	The previous value of A is displayed.
2A	CLR	Press CLR once to clear LED value. Press CLR one more time to exit the calibration mode.		
3	5.0	New value may be entered.		5.0 flashes
4	ENTER	If the value for A is acceptable, it may be entered. Upon completion of calculation, next message appears.	A Calib in progress Buffer B in cell ENTER grad value	The previous value of B is displayed.
4A	CLR	Press CLR once to clear LED value. Press CLR one more time to exit the calibration mode.		
5	95.0	If new value for B is required, it may be entered.	B calib	95.0 flashes. in progress
6	ENTER	0 will correspond to 0 volt output on chart recorder; 100 will correspond to 100% full scale output. 95 will deflect the chart pen 95% full scale.	CALIB values are A=5.0 B=95.0	Gradient percent in cell at time.

2.7 RS-232-C Control of the Model 1710 Gradient Monitor

The Model 1710 Gradient Monitor can be completely controlled through the RS-232 port on the rear panel. The Model 1710 monitor is configured as data terminal equipment according to EIA RS-232 standards. To connect this to a personal computer will require a null modem cable.

The following connection is recommended for controlling the Model 1710 Gradient Monitor through the remote terminal.

Computer (DTE) signal name & pin	Model 1710 (DTE) signal name & pin
-------------------------------------	---------------------------------------

TXD 3-----	RXD 2
RXD 2-----	TXD 3
RTS 4-----	CTS 5
CTS 5-----	RTS 4
DSR 6-----	DTR 20
DTR 20-----	DSR 6
GND 1-----	GND 1
GND 7-----	GND 7

All commands are ASCII characters.

Command syntax: <device name><space><command>[<space><command args>]

Device name: GRAD.MONITOR

Appendix I contains all commands supported through the RS-232 port.

Note: A detection cell must be connected for proper operation of the RS-232 port.

2.8 Care of Flow Cells and Dip Probe

After use, the flow cell should be flushed with deionized water to remove salts that may crystalize in the flowpath. To store the flow cell for more than a few days, replace the water with isopropyl alcohol, and cap the tube ends. Remember that direct addition of isopropyl alcohol to a salt solution or buffer may cause precipitation to occur. Always rinse the flow cell with water prior to the introduction of aqueous buffers.

The dip probe should be flushed with deionized water, too, irrigating the detection surfaces. Do not leave the dip probe in prolonged contact with high ionic strength solutions. The dip probe may be stored dry.

Section 3 Troubleshooting Guide

Problem	Possible Cause	Solution
1. Display screens do not appear.	Power is not supplied to the instrument.	Check power cord connections and power strip switches.
	Current has exceeded 1 amp, and fuses are blown.	Replace 1 amp fuses.
	Failure on board 152.	Contact Instrument Service at Bio-Rad.
2. Keypad is locked up.	An external cell is not connected to the instrument.	Connect an external cell to the cell port on the back panel.

Problem	Possible Cause	Solution
	Failure on board 151.	Contact Instrument Service at Bio-Rad.
3. Neither peaks nor gradient profile appear on output device.	Signal cable not connected at monitor or device terminals.	Connect signal cable to correct terminals.
	Data channel not selected on multi-channel data stations.	Select correct data channel on multi-channel data stations.
	Cell flow path not connected to chromatograph.	Connect inlet tubing of cell to outlet of column, or first monitor if dual detection system.
	Eluant not flowing.	Check eluant reservoir level and pump flow rate. Prime or reset pump if necessary.
4. Smaller than expected peaks or gradient profile.	Output to integration device is set at 10 mV.	Set signal output switch to 1 V.
	Integration device scaling is misset.	Change integration device vertical scaling or attenuation.
	Leak in fittings.	Check all fittings for leakage.
	Fewer electrolytes in solution.	Verify sample loop volume, sample concentration, or buffer B concentration.
	If in CONC or GRAD modes, calibration values for previous buffers may be in use.	Recalibrate with current buffers.
5. Larger than expected peaks or gradient profile.	Output to chart recorder is set at 1 V.	Set signal output switch to 10 m V.
	Chart recorder input voltage is misset.	Change chart recorder input voltage to 10 mV.
	If in CONC or GRAD modes, calibration values for previous buffers may be in use.	Recalibrate with current buffers.
6. Spikes appear in baseline or gradient profile.	Air bubble passing through flow cell.	Degas eluant. Check for loose plumbing connections. Elevate flow cell outlet to release bubble.

Problem	Possible Cause	Solution
7. Ghost peaks appear in gradient profile.	Charged sample compound (e.g., protein) detected.	No problem.
	Air bubble passing through flow cell.	Degas eluant. Check for loose plumbing connections. Elevate flow cell outlet to release bubble.
8. Noisy baseline.	Air bubble passing through flow cell.	Degas eluant. Check for loose plumbing connections. Elevate flow cell outlet to release bubble.
	Pump pulsation.	Reprime pump heads. Consider installation of a pulse damper.
	Column unequilibrated or dirty.	Remove, replace, or clean column. Use clean eluant.
	Flow cell dirty.	Clean flow cell with 0.1 N nitric acid.
	Electrical noise.	Remove instruments (especially those with motors) that share the same power strip and circuit.
9. Drifting baseline.	Column and tubing not thermally insulated.	Insulate tubing. Use column heater.
	Environmental temperature out of 5 to 40° C range.	Modify environmental temperature.
	Temperature coefficient of resistance differs from 2% per 1° C.	Select different eluant.
	Column unequilibrated or dirty.	Remove, replace, or clean column. Use clean eluant.
	Flow cell dirty.	Clean flow cell with 0.1 N nitric acid.
10. High salt standard curve deviates from linearity.	See Section 1.2.	Use standard curve based on conductivity values to determine concentration values. Do not use CONC mode.

Section 4: Ordering Information

<u>Catalog Number</u>	<u>Product Description</u>
167-0430	Model 1710 Gradient Monitor, 100/120 V
167-0431	Model 1710 Gradient Monitor, 220/240 V
167-0434	Standard Flow Cell for 1710
167-0435	High Sensitivity Flow Cell for 1710
167-0436	Dip Probe for 1710
167-0445	Conductivity Standard

Appendix I

Gradient Monitor Commands and Responses

SYNTAX:

<device name><space><commands> [<space><command args.>] <cr>

device name: "GRAD.MONITOR" - GRADIENT MONITOR DETECTOR

<command>:	"ID"	- request device ID. Gradient Monitor id; no arguments
<response>:	"CM E" <space><error code><space><id><cr>	
<id>:	"0330"	gradient monitor id code
<command>:	"RESET"	- reset device to power up configuration; no arguments
<response>:	"CM E" <space><error code><cr>	
<command>:	"CLEAR"	- exit from calibration entry mode; no arguments - restore old calibration point values
<response>:	"CM E" <space><error code><cr>	
<command>:	"MODE"	- set display mode; argument md
<cmd args>:	md	- mode string
	"COND"	conductance mode
	"CONC"	concentration mode
	"GRAD"	gradient mode
<response>:	"CM E" <space><error code><cr>	
<command>:	"RANGE"	- set device range; argument rn
<cmd args>:	rn	- range number
	"1"	0.000 to +/- 1.999 μ S
	"2"	0.00 to +/- 19.99 μ S
	"3"	0.0 to +/- 199.9 μ S
	"4"	0 to +/- 1999 μ S
	"5"	0.00 to +/- 19.99 mS
<response>:	"CM E" <space><error code><cr>	
<command>:	"CALIB"	- sets a calibration point's conc. or % grad.; arguments ty, et, cv. The "CALIB" command specifies the calibration eluant A or B and its conc. or % grad. value depending on the calibration type argument.
<cmd args>:	ty	- calibration type, one of:
	"CONC"	specifies concentration calibration
	"GRAD"	specifies gradient calibration
<cmd args>:	et	- Eluant (A or B)
	"A"	specifies calibration of eluant A
	"B"	specifies calibration of eluant B
<cmd args>:	cv	- Calibration value for eluant (A or B) The calibration value is in ASCII encoded floating point e format with a range of 0.000 to 1999 while in CONC mode and a range of 0.0 to 100.0 while in GRAD mode.
<response>:	"CM E <error code>"	

<command>: "CCALIB" - sets a calibration point's conc. or % grad. and conductance; arguments ty, et, cv, cd
The "CCALIB" command specifies the calibration eluant A or B, its conc. or % grad. value depending on the calibration type, and its conductance value for the eluant. The command sets both the calibration value and conductance value of the eluant.

<cmd args>: ty - calibration type, one of:
"CONC" specifies concentration calibration
"GRAD" specifies gradient calibration

<cmd args>: et - Eluant (A or B)
"A" specifies calibration of eluant A
"B" specifies calibration of eluant B

<cmd args>: cv - Calibration value for eluant (A or B)
The calibration value is in ASCII encoded floating point E format with a range of 0.0000 to 1999 for a CONC calibration and a range of 0.0 to 100.0 for a GRAD calibration.

<cmd args>: cd - Conductivity value for eluant (A or B)
Conductivity is value in ASCII encoded floating point E format. Conductivity values are in units of Siemens. The range of conductivity values is -1.999E-6 to 1.999E-2 Siemens.

<response>: "CM E" <space><error code><cr>

<command>: "STATUS" - request a status packet; no arguments

<response>: A multi-record status report, fields separated by space characters, and each line terminated by a <CR>:
"DEV <mode><range><calib>"
"CPT CONC A <calib val A><cond val A>"
"CPT CONC B <calib val B><cond val B>"
"CPT GRAD A <calib val A><cond val A>"
"CPT GRAD B <calib val B><cond val B>"
"CUR <conductivity><conc.><%gradient>"
"CM E <error code>"

<mode>: Display mode, one of: "COND", "CONC", or "GRAD", indicating the mode of the front panel display.

<range>: Instrument range number:
"1" 0.000 to +/- 1.999 μ S
"2" 0.00 to +/- 19.99 μ S
"3" 0.0 to +/- 199.9 μ S
"4" 0 to +/- 1999 μ S
"5" 0.00 to +/- 19.99 mS

<calib>: Keyword "CALIB" signifies that the instrument is in a calibration sequence. Otherwise, "NOCAL" is listed here.

<calib val A>: Calibration value for calibration point A. The calibration value is in ASCII encoded floating point e format with a range of 0.000 to 1999 for CONC points and a range of 0.0 to 100.0 for GRAD points. If the instrument is in a calibration sequence, the old calibration value for the point is returned in this field.

<cond val A>: Conductivity value for calibration point A. The conductivity value is in ASCII encoded floating point E format. Conductivity values are represented in units of Siemens. The range of a conductivity values is -1.999E-6 to 1.999E-2 Siemens. If the instrument is in a calibration sequence, the old conductivity value for the point is returned in this field.

<calib val B>: Calibration value for calibration point B. The calibration value is in ASCII encoded floating point E format with a range of 0.0000 to 1.999 for CONC points and a range of 0.0 to 100.0 for GRAD points. Conductivity values are represented in units of Siemens. The range of a conductivity values is -1.999E-6 to 1.999E-2 Siemens. If the instrument is in a calibration sequence, the old calibration value for the point is returned in this field.

<cond val B>: Conductivity value for calibration point B. The conductivity value is in ASCII encoded floating point E format. Conductivity values are represented in units of Siemens. The range

of a conductivity values is -1.999E-6 to 1.999E-2 Siemens. If the instrument is in a calibration sequence, the old calibration value for the point is returned in this field.

<conductivity>: Current conductivity reading value. The conductivity value is in ASCII encoded floating point E format. Conductivity values are represented in units of Siemens. The range of a conductivity values is - 1.999E-6 to 1.999E-2 Siemens.

<%gradient> The current % gradient value.

<conc.>: The current concentration value.

<command>: "+STATUS" - request status at 3 sec intervals; no arguments

<response>: "CM E <error code><cr>"

The status message is in the following format:

"CUR <conductivity><conc.><%gradient>"

- At 3 second intervals, this status message will be transmitted from the gradient monitor.

- While the "+STATUS" command is active the "CCALIB" command may not be executed from the host. All other commands may be executed from the host.

- Status transmission is delayed until an incoming command from the host is executed and its response transmitted.

- While the "+STATUS" command is active, the gradient monitor will continue sending event messages.

<command>: "-STATUS" - Halts status transmission; no arguments

<response>: "CM E <error code><cr>"

Messages Initiated by the Gradient Monitor

- When an event, such as a display mode change, the entering of new calibration points, or a range change occurs, the gradient monitor will transmit a response message in the following format and syntax:

<response>: "EV E" <space> <error code> <cr>

Gradient Monitor Error Codes

1. Use the code 0000 for successful completion.
2. All completion codes are 16 bit integers expressed in unsigned hexadecimal.
3. All codes describing an error condition have the high bit set (bit 15).

0000	- No error
8081	- Invalid Command
8082	- Parameter Out of Range
8083	- Device Not Ready (3 minute warm up period)
8084	- Device not in Calibration Mode
8085	- Device in Calibration Mode
8086	- Hardware error